

Part 1: Overview of Time Series

Text references: §12.1–12.3 in Ruppert & Matteson

This course is about time series analysis, with emphasis on techniques and models useful in analysis of financial time series.

The topics to be covered include the following (subject to change):

- Motivation
- Basic Definitions
- The ARIMA Models
- GARCH Models
- Stochastic Volatility Models
- Multivariate Time Series Models
- Nonparametric Time Series Models
- Regression with Correlated Errors
- . . .

We will discuss how to implement these procedures in R.

1

2 Definition of a Time Series

A *stochastic process* is a collection of random variables

$$\{X_t(\omega), t \in \mathcal{T}\}$$

3 defined on a probability space $(\Omega, \mathcal{F}, P)$.

4 A *realization* of a stochastic process is a function $X_t(\omega)$ of t (holding
5 $\omega \in \Omega$ fixed). In particular, this can be regarded as a set of numbers
6 indexed by t .

A *time series* is simply stochastic process for which $\mathcal{T}$ is a set of points in time. In this course, we will assume that

$$\mathcal{T} = \{0, \pm 1, \pm 2, \dots\},$$

7 that is, the time series is a set of random variables indexed by integer
8 valued times. Typically, however, we will only observe a portion of
9 these, from time $1, 2, \dots, n$.

1 Note also that people tend to use the phrase “time series” to refer to
2 either the stochastic process itself (which is a collection of random
3 variables), or its realization (which is a collection of numbers).

4 For example, Figure 1 shows daily closing prices of SBUX shares (in
5 dollars) over all of 2016 (252 trading days). For this particular time
6 series, we have observed values at times $1, 2, \dots, 252$. In general, the
7 time unit is arbitrary, but it is most often assumed that observations
8 take place at equally-spaced points in time.

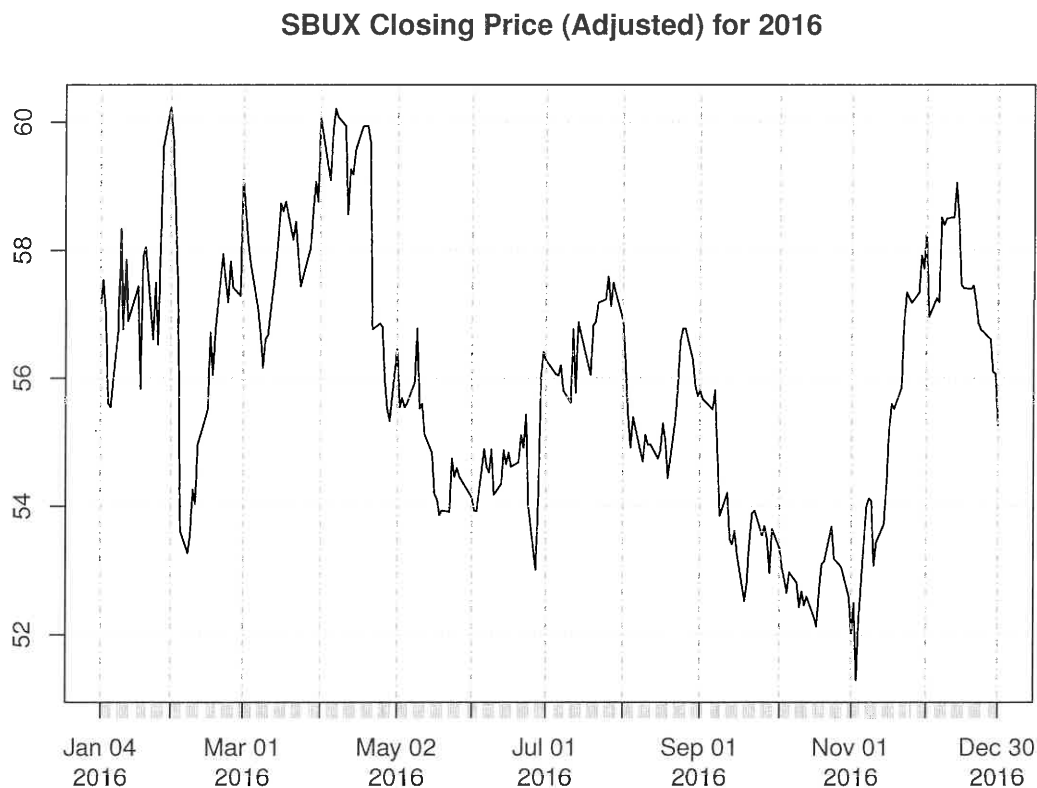


Figure 1: Time series of the closing price (adjusted) for SBUX for 2016.

1 The following sequence of commands created Figure 1.

```
2 library(quantmod)
3
4 SBUX = getSymbols("SBUX", from="2016-1-1",
5                   to="2016-12-31", auto.assign=F)
6
7 plot(Ad(SBUX),
8      main="SBUX Closing Price (Adjusted) for 2016")
```

9 Note how the columns of SBUX have a specific form that leads the
10 `plot()` function to handle it in a specific way. Such a function is
11 called a *generic function* in R. `predict()` is another example of a
12 generic function.

1

2 **Typical Goals of a Time Series Analysis**

3 Typically, there are two main things we might want to do in time
4 series analysis.

5 The first is to generate forecasts of future values (or rather, predictive
6 distributions for future values), based on current and past observa-
7 tions.

8 The second is to gain some understanding of the process itself.

1 Example: Forecasting for Option Pricing

2 One often thinks of forecasting as predicting the “most likely” value
3 at some time in the future, based on what we have seen up to the
4 current time. Obtaining such a “point forecast” is generally **not** par-
5 ticularly useful, because it is typically not accurate.

6 Consider, for example, the problem of option pricing. Suppose that
7 a particular share price has behaved as shown in Figure 2.

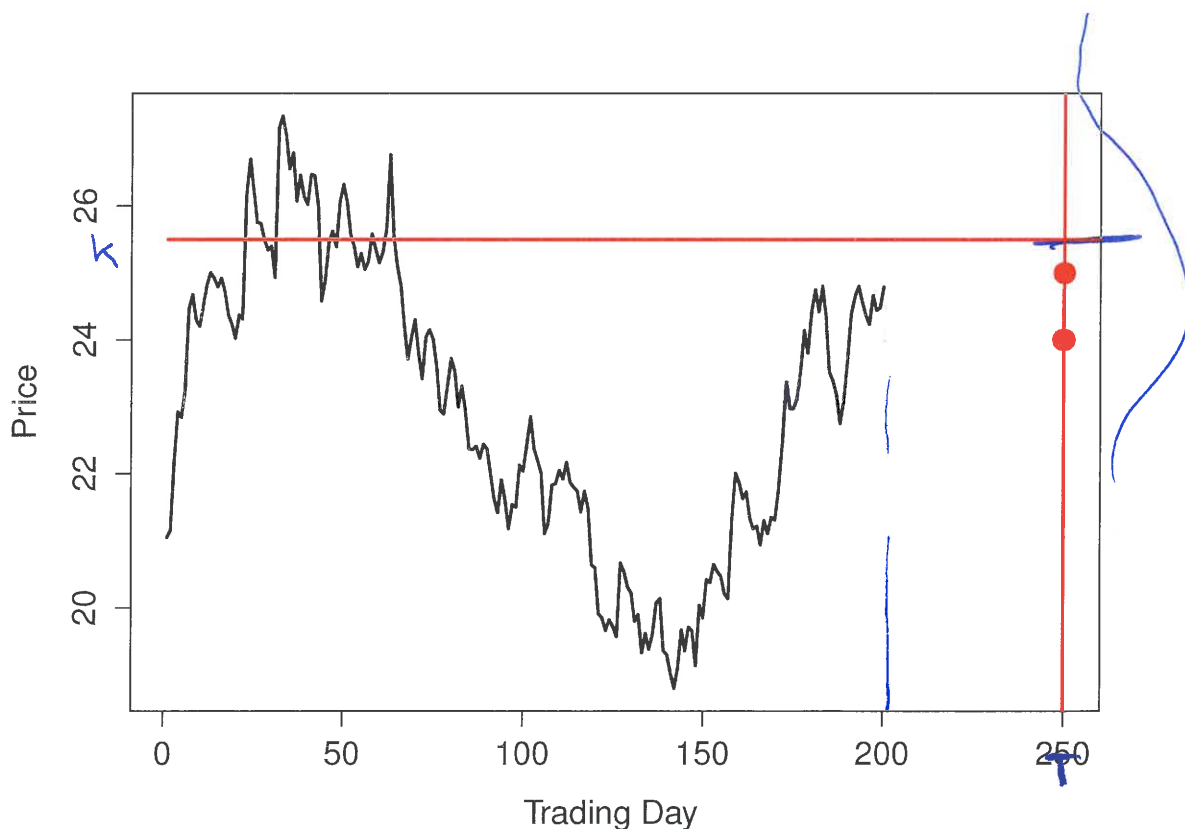


Figure 2: Time series with option expiration and strike price included.

1 Suppose that you can buy a call option on this share, with expiry
2 time T and strike price K . What should the option be worth? If
3 we were to make point forecasts, two seemingly reasonable choices
4 would be as indicated by the circles (1) and (2) in the picture. Clearly,
5 for each of these outcomes, the value of the call option would be zero
6 at expiry (since the share price for these predictions is less than the
7 strike price). However, it would be a bad idea to sell someone such
8 options for a price of zero.

9 What we really need here in order to make sense of the valuation
10 problem is the **distribution** of the share price at time T , given what
11 we have observed up to the current time.

12 This is often referred to as a *predictive distribution*.

1 Example: Constructing Indicators

2 How can a time series consisting of past information (e.g., price) be
3 “transformed” into one or more indicators that can be useful for fore-
4 casting future market behavior?

5 Standard time series models fit a process to the series, requiring the
6 estimation of a few parameters.

7 In this way, time series models are identical to statistical models con-
8 sidered in previous courses: they replace the real data generating
9 process with a much simpler process, characterized only by a few
10 unknown constants and some distributional assumptions.

- 1 **Exercise:** Two time series are shown in Figure 3. One is the daily
2 returns for a particular stock for 2013; the other is simulated from
3 a simple time series model (the AR(1) model) fit to those real data.
4 Which is which?

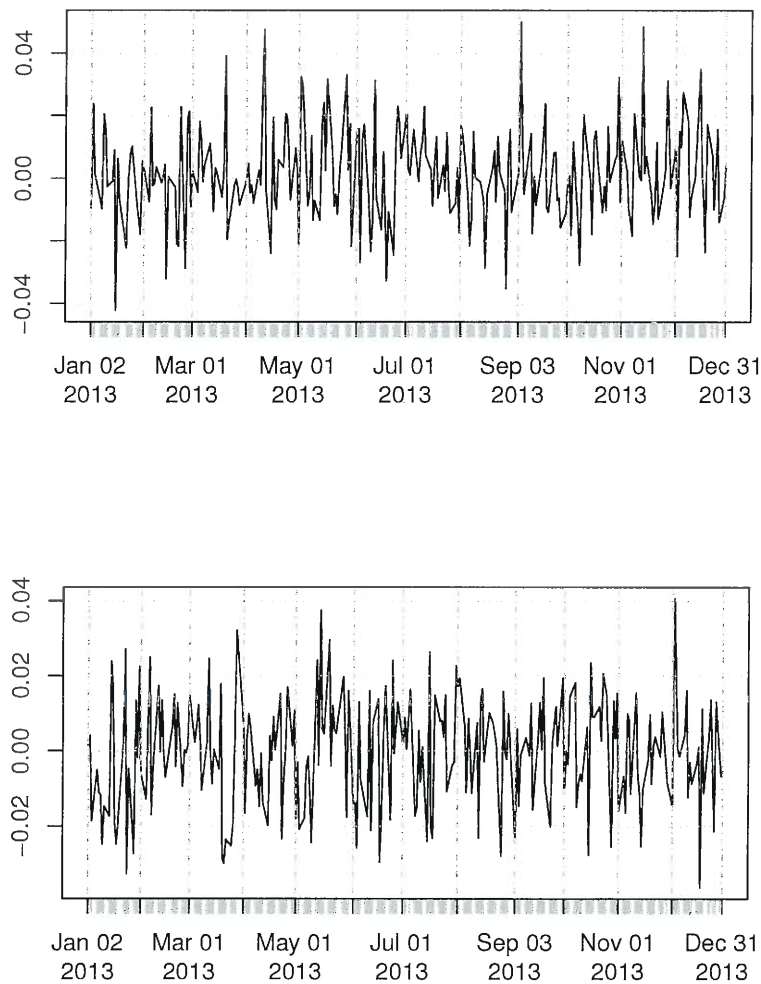


Figure 3: Two time series. One is the real daily returns for a stock for 2013, the other is simulated from a time series model fit to those data.

A Standard Approach to Time Series Modelling

A standard approach to time series modelling is as follows.

1. Plot the time series. There are a lot of features that are still most readily identified by a human rather than a computer.
2. Transform the original time series to a stationary time series:
 - (a) Determine whether or not to apply a Box-Cox (log) transform.
 - (b) Check for trend, seasonal component, sharp changes, outliers, etc., and then carry out appropriate transformations.
3. Fit a model to the stationary time series.
4. Carry out diagnostic checks to ensure that the fitted model is reasonable. If necessary, go back to Step 2.
5. Generate forecasts (i.e. predictive distributions) for the model.
6. Invert the transformations of Step 2, in order to obtain forecasts for the original time series.

Stationarity

An important property of many time series models is *stationarity*. A large number of methods have been developed to work with stationary time series. Although most time series are not initially stationary, they can typically be transformed into a time series that is approximately stationary.

To define (weak) stationarity, we need to first introduce the *autocovariance function* for a time series $\{X_t\}$, defined by

$$\begin{aligned}\gamma_X(r, s) &= \text{Cov}(X_r, X_s) \\ &= E[(X_r - EX_r)(X_s - EX_s)] \\ &= E(X_r X_s) - EX_r EX_s\end{aligned}$$

The time series $\{X_t\}$ is then said to be (*weakly*) *stationary* if

1. $EX_t = \mu$ for all t , (μ doesn't depend on t), $E(X_t | X_{t-1}) \neq \mu$
2. $\gamma_X(t, t+h) = \gamma_X(s, s+h)$ for all integers t, s, h .

(Implicit in these conditions is the assumption that the first two moments of X_t are finite.)

- 1 The second property implies that the autocovariance function de-
2 pends on only the lag h between time points This is equivalent to
3 requiring that $\gamma_X(t, t + h)$ is not a function of t .

Thus if a process is stationary, we usually write

$$\gamma_X(h) = \text{Cov}(X_t, X_{t+h}).$$

- 4 (This is valid since for a stationary process, the term on the right
5 hand doesn't depend on t .)

- 6 **Exercise:** Explain why, if $\{X_t\}$ is stationary, then $V(X_t)$ is not a func-
7 tion of t .

8 $V(X_t) = \text{Cov}(X_t, X_t) = \gamma_X(0)$
9 i.e., not a function of t

10 $V(X_t | X_{t-1} = a) \neq V(X_t)$
11 GARCH
12
13
14
15

1 **Exercise:** Review the concepts and properties of a σ -field $\mathcal{F}_t$.

2 $\mathcal{F}_t$ represents all of the information available
3 at time t

4 Saying that " X_t is $\mathcal{F}_t$ -measurable" is
5 saying I know X_t at time t
6

7 Properties:

8 (1) $\mathcal{F}_s \subseteq \mathcal{F}_t \quad s \leq t$
9

10 (2) $E(X) = E(E(X|\mathcal{F}_t))$ for any X
11

12 (3) $E(X|\mathcal{F}_t) = X$ if X is $\mathcal{F}_t$ -measurable
13

14 (4) $E(XY|\mathcal{F}_t) = X E(Y|\mathcal{F}_t)$ if X is $\mathcal{F}_t$ -meas
15

16 (5) $E(E(X|\mathcal{F}_t)|\mathcal{F}_s) = E(E(X|\mathcal{F}_s)|\mathcal{F}_t)$
17 $= E(X|\mathcal{F}_s)$ when $s \leq t$
18

19 (6) If $s \leq t$, then $E(X|\mathcal{F}_s)$ is $\mathcal{F}_t$ -measurable
20

1 Example: The White Noise Process

Suppose that $\{X_t\}$ is a sequence of random variables with

$$\gamma_X(r, s) = \begin{cases} \sigma^2, & r = s, \\ 0, & \text{otherwise,} \end{cases}$$

2 and $EX_t = \mu$.

Then we say that $\{X_t\}$ is a (*weak*) *white noise process* with mean μ and variance σ^2 , and denote this by

$$\{X_t\} \sim \text{WN}(\mu, \sigma^2).$$

3 If the mean is not specified, then it is assumed that $\mu = 0$.

4 *IID white noise* is a special case of white noise in which the X_t are iid.

5 **Exercise:** Show that the white noise process is stationary.

6 ① $E(X_t) = \mu$ for all t is given

7 ② $\gamma_X(t, t+h) = \begin{cases} \sigma^2, & h=0 \\ 0, & \text{otherwise} \end{cases}$ for all t

10
11 So, $\{X_t\}$ is stationary

1 Example: Martingale Difference Series/Sequence (MDS)

- 2 An important intermediate case between weak white noise and IID
 3 white noise is the *(homoskedastic) martingale difference series (MDS)*.

We say that $\{X_t\}$ is a MDS if X_t is $\mathcal{F}_t$ measurable and

$$E(X_t | \mathcal{F}_{t-1}) = 0$$

- 4 for all t . In addition, we say that an MDS is *homoskedastic* if $V(X_t) =$
 5 σ^2 for all t .

- 6 **Exercise:** Show that a homoskedastic MDS is weak white noise with
 7 mean zero.

8 (1) $E(X_t) = E(E(X_t | \mathcal{F}_{t-1})) = E(0) = 0$

9 (2) Note that $V(X_t) = \sigma^2$ for all t (given)

10 Assume $r < s$

11
$$\begin{aligned} \text{Cov}(X_r, X_s) &= E(X_r X_s) \\ 12 &= E(E(X_r X_s | \mathcal{F}_r)) \\ 13 &= E(X_r E(X_s | \mathcal{F}_r)) \\ 14 &= E(X_r E(E(X_s | \mathcal{F}_{s-1}) | \mathcal{F}_r)) \end{aligned}$$

15
$$= 0$$

16
$$= 0$$

since $\mathcal{F}_r \subseteq \mathcal{F}_{s-1}$

Conclude that $\{X_t\}$ is (weak) white noise, mean 0

- 1 **Exercise:** What is the relationship between (1) IID white noise, (2)
2 homoskedastic MDS, and (3) weak white noise?

IID white noise, mean 0



homoskedastic MDS



weak white noise,
mean 0

Example of series which is white
noise but not MDS

$$X_t = \epsilon_t + \epsilon_{t-1} \epsilon_{t-2} \quad \{\epsilon_t\} \text{ iid white noise, mean 0}$$

$$E(X_t | \mathcal{F}_{t-1}) = \epsilon_{t-1} \epsilon_{t-2} \neq 0$$

Example: The Random Walk

Suppose that X_t is defined by

$$X_t = \sum_{s=1}^t Z_s, \quad \{Z_s\} \text{ is an homoskedastic MDS,}$$

for $t = 1, 2, 3, \dots$. This is called a *random walk*.

Exercise: Is this random walk stationary?

$$\textcircled{1} E(X_t) = \sum_{s=1}^t E(Z_s) = 0$$

$$\textcircled{2} \gamma_X(t, t+h) = E(X_t X_{t+h}), \quad h \geq 0$$

$$= E(E(X_t X_{t+h} | \mathcal{F}_t))$$

$$= E(X_t E(X_{t+h} | \mathcal{F}_t))$$

$$= E(X_t X_t)$$

$$= V(X_t)$$

$$= V\left(\sum_{s=1}^t Z_s\right) = t\sigma^2$$

$\gamma_X(t, t+h)$ depends on t , so not stationary

Example: The MA(1) Process

Suppose that

$$X_t = Z_t + \theta Z_{t-1}, \quad \{Z_t\} \sim \text{WN}(0, \sigma^2).$$

for some constant θ . This is called the *moving average of order one* (MA(1)) process.

Exercise: Is the MA(1) process stationary?

$$\textcircled{1} E(X_t) = E(Z_t + \theta Z_{t-1}) = E(Z_t) + \theta E(Z_{t-1}) = 0$$

$$\textcircled{2} \text{Cov}(X_t, X_{t+h}) = E(X_t X_{t+h})$$

$$= E((Z_t + \theta Z_{t-1})(Z_{t+h} + \theta Z_{t+h-1}))$$

$$= E(Z_t Z_{t+h}) + \theta E(Z_{t-1} Z_{t+h}) + \theta E(Z_t Z_{t+h-1}) + \theta^2 E(Z_{t-1} Z_{t+h-1})$$

know that $E(Z_t Z_s) = \begin{cases} \sigma^2, & t=s \\ 0, & t \neq s \end{cases}$

$$= \begin{cases} E(Z_t^2) + \Theta^2 E(Z_{t+1}^2) & , h=0 \\ \Theta E(Z_t^2) & , h=1 \\ \Theta E(Z_{t+1}^2) & , h=-1 \\ 0 & \text{otherwise} \end{cases}$$

$$= \begin{cases} \sigma^2 + \Theta^2 \sigma^2 & , h=0 \\ \Theta \sigma^2 & , |h|=1 \\ 0 & , \text{otherwise} \end{cases}$$

does not depend on t , so conclude that the series is stationary

1 The ARCH(1) Process

2 Suppose that

$$3 \quad X_t = Z_t \sqrt{\omega + \alpha X_{t-1}^2}, \quad \{Z_t\} \text{ is IID white noise, mean } 0, \text{ variance } 1$$

4 This is an *autoregressive conditional heteroskedastic of order one (ARCH(1))*

5 process. Assume that $\omega > 0$ and $0 \leq \alpha < 1$. Further assume that

6 $V(X_t) < \infty$ for all t .

7 **Exercise:** Is this ARCH(1) process stationary?

$$8 \quad \textcircled{1} \quad E(X_t) = E\left(Z_t \sqrt{\omega + \alpha X_{t-1}^2}\right) \\ 9 \quad = E(Z_t) E\left(\sqrt{\omega + \alpha X_{t-1}^2}\right) \quad \text{since } Z_t \text{ independent} \\ 10 \quad = 0 \quad \text{of } \sqrt{\omega + \alpha X_{t-1}^2}$$

$$11 \quad \textcircled{2} \quad \gamma_X(t, t+h) = E(X_t X_{t+h}) \\ 12 \quad = E\left(\left(Z_t \sqrt{\omega + \alpha X_{t-1}^2}\right) \left(Z_{t+h} \sqrt{\omega + \alpha X_{t+h-1}^2}\right)\right) \\ 13$$

14 Assume $h \neq 0$

15 If $h > 0$, Z_{t+h} ind. of $Z_t \sqrt{\omega + \alpha X_{t-1}^2} \sqrt{\omega + \alpha X_{t+h-1}^2}$
 16 $h < 0$, Z_t ind. of $Z_{t+h} \sqrt{\omega + \alpha X_{t-1}^2} \sqrt{\omega + \alpha X_{t+h-1}^2}$

17 In either case, $E(X_t, X_{t+h}) = 0$, not
 18 a function of t
 19

when $h=0$,

$$\gamma_X(t, t+h) = V(X_t) = E(X_t^2)$$

$$= E(Z_t^2 (\omega + \alpha X_{t-1}^2))$$

$$= E(Z_t^2) E(\omega + \alpha X_{t-1}^2)$$

$$= \omega + \alpha E(X_{t-1}^2)$$

$$= \omega + \alpha V(X_{t-1})$$

$$= \omega + \alpha (\omega + \alpha V(X_{t-2}))$$

$$= \sum_{i=0}^T \omega \alpha^i + \alpha^{T+1} V(X_{t-T-1})$$

$$\text{as } T \rightarrow \infty \quad = \frac{\omega}{1-\alpha} \quad \text{since } 0 \leq \alpha < 1$$

So, the series $\{X_t\}$ is stationary

Also note that $\{X_t\}$ is white noise

In fact, $\{X_t\}$ is MDS

$$\begin{aligned} E(X_t | \mathcal{F}_{t-1}) &= E(Z_t \sqrt{\omega + \alpha X_{t-1}^2} | \mathcal{F}_{t-1}) \\ &= \sqrt{\omega + \alpha X_{t-1}^2} E(Z_t | \mathcal{F}_{t-1}) \\ &= 0 \end{aligned}$$

What do Stationary Processes Look Like?

Keep in mind that stationarity is a property of **the time series process that generated the observed data** not a property of **the observations themselves**.

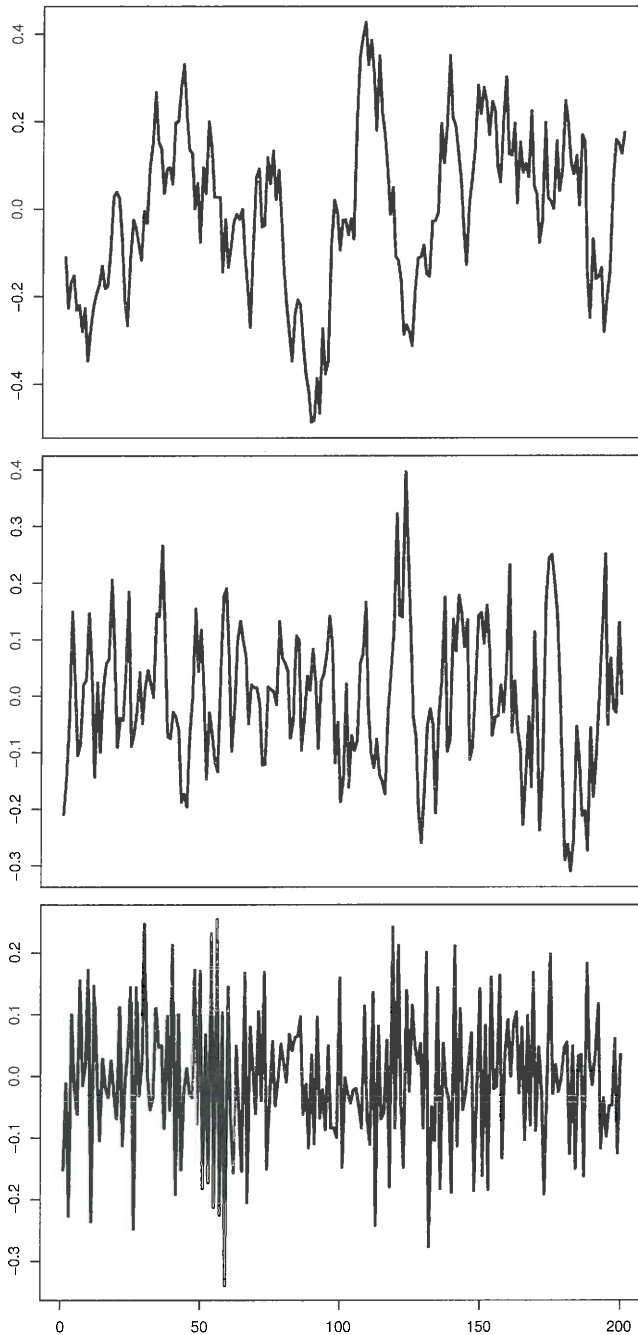
So, it is not really possible to determine if the process is stationary by looking at a realization (observed data).

Fortunately, many observed data series are sufficiently long that they give us a good look at the overall behavior of the process.

In particular, stationary series exhibit *mean reversion*, which means that the average level of the series tends to return to fluctuate around the overall mean μ .

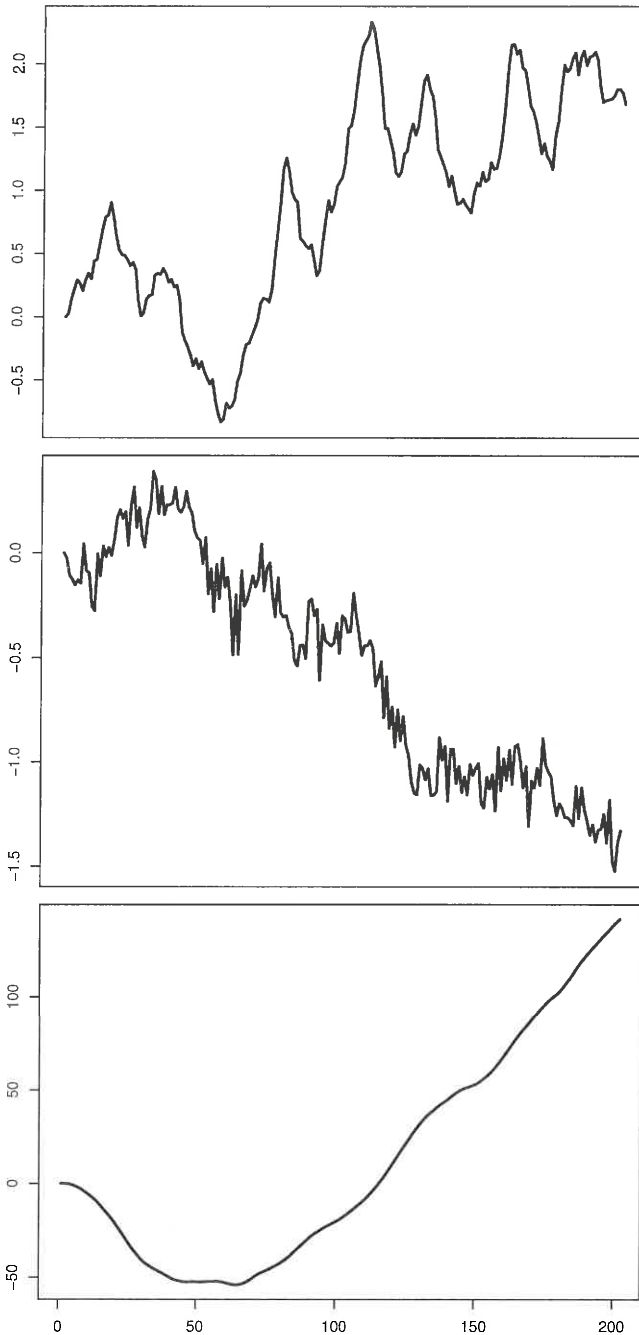
Likewise, while the variance of the process may vary over time, it should tend to fluctuate around σ^2 .

Of course, no real process is exactly stationary. As is often the case, it is an question of whether or not the approximation is sufficiently valid for the model to be useful.

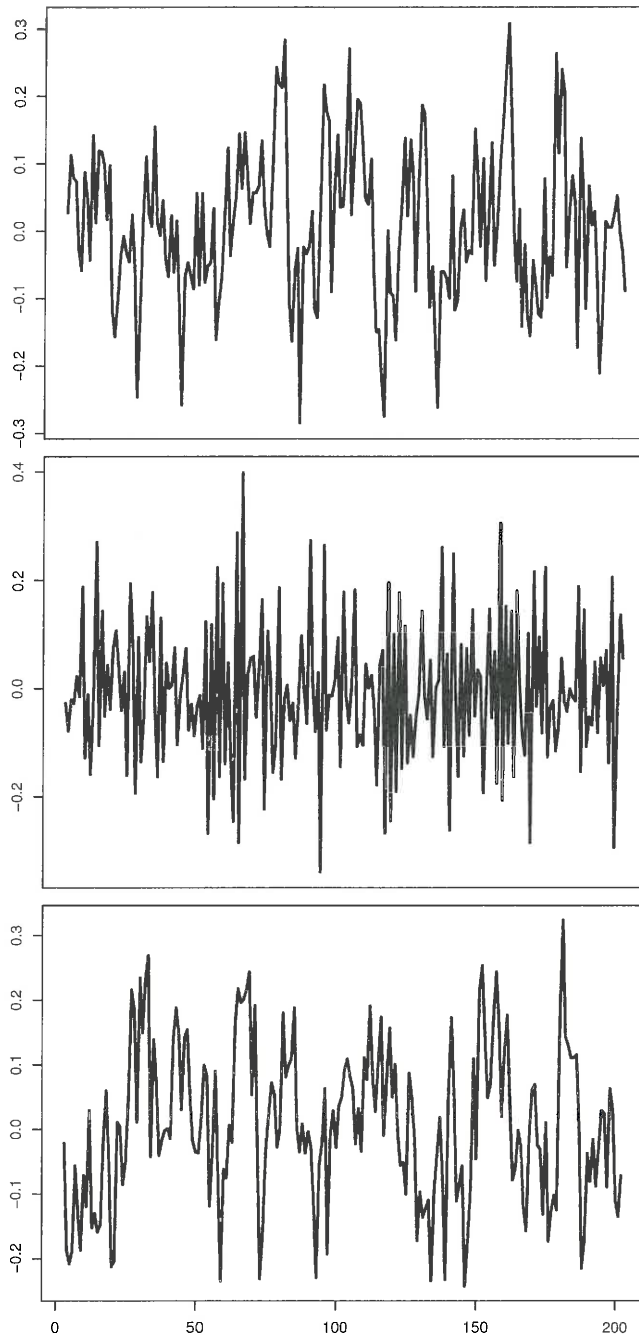


Generated from
stationary AR(1)
model

$$X_t = \alpha + \beta X_{t-1} + \epsilon_t$$



not stationary
lack of mean
reversion



All stationary

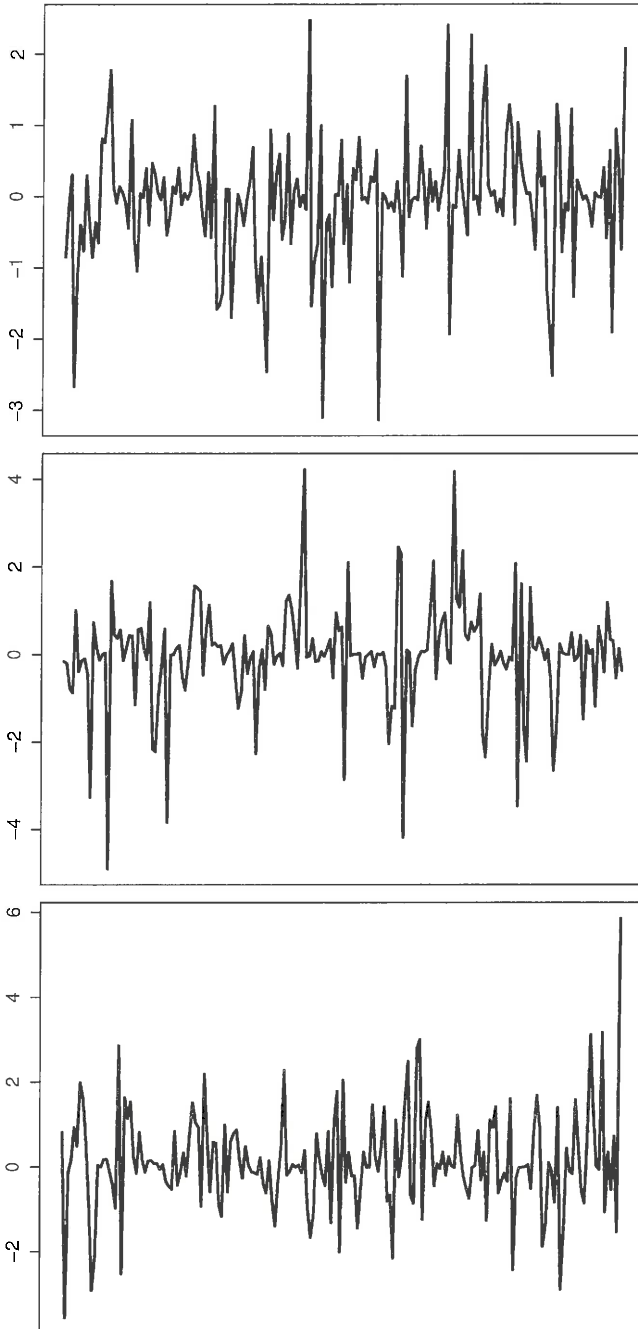
← formed by taking first differences of the two series on previous page

$$Y_t = X_t - X_{t-1}$$

This is the "second difference" of the third series on previous page

$$Y_t = (X_t - X_{t-1}) - (X_{t-1} - X_{t-2})$$

$$\neq X_t - X_{t-2}$$



stationary, from
 $ARCH(1)$

1

2 **Strict Stationarity**

3 There is a different form of stationarity, called *strict stationarity*, which
4 is not as important in practice because of its restrictiveness.

5 The process $\{X_t\}$ is said to *strictly stationary* if the joint distributions
6 of $(X_{t_1}, \dots, X_{t_k})$ and $(X_{t_1+h}, \dots, X_{t_k+h})$ are the same, for all positive
7 integers k , and for all possible choices $t_1, t_2, \dots, t_k, h \in \mathbb{Z}$.

8 Note that if a process is Gaussian, then weak stationarity implies
9 strict stationarity.

The Autocovariance and Related Functions

For a stationary time series, recall that $\gamma_X(h) = \text{Cov}(X_t, X_{t+h})$.

The autocovariance function γ has the following properties.

1. $\gamma_X(0) \geq 0$

2. $|\gamma_X(h)| \leq \gamma(0)$

3. $\gamma_X(h) = \gamma_X(-h)$

Exercise: Prove each of the above properties.

$$\textcircled{1} \quad \gamma_X(0) = V(X_t) \geq 0$$

$$\textcircled{2} \quad \text{Recall: } \text{Corr}(X_t, X_{t+h}) = \frac{\text{Cov}(X_t, X_{t+h})}{\sqrt{V(X_t)V(X_{t+h})}} = \frac{\gamma_X(h)}{\gamma_X(0)}$$

$$\gamma_X(h) = \text{Corr}(X_t, X_{t+h}) \gamma_X(0)$$

$$\begin{aligned} |\gamma_X(h)| &= |\text{Corr}(X_t, X_{t+h}) \gamma_X(0)| \\ &= |\text{Corr}(X_t, X_{t+h})| \gamma_X(0) \\ &\leq \gamma_X(0) \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad \gamma_X(h) &= \text{Cov}(X_t, X_{t+h}) = \text{Cov}(X_{t-h}, X_t) \\ &= \text{Cov}(X_t, X_{t-h}) = \gamma_X(-h) \end{aligned}$$

The *autocorrelation function* is the normalized version of the autocovariance function, given for a process $\{X_t\}$ by

$$\rho_X(h) = \frac{\gamma_X(h)}{\gamma_X(0)}, \quad h = 0, 1, 2, \dots$$

This has same properties as the autocovariance function, but note that by construction $\rho_X(0) = 1$.

Sample Autocovariance and Autocorrelation

Note that the autocovariance and autocorrelation functions are properties of a *model*. In practice, we will have to use data to estimate these functions.

This estimate is called the *sample autocovariance function*

$$\hat{\gamma}_X(h) = \frac{1}{n} \sum_{j=1}^{n-h} (x_{j+h} - \bar{x})(x_j - \bar{x}), \quad 0 \leq h < n,$$

where $\bar{x} = \frac{1}{n} \sum_{j=1}^n x_j$.

The *sample autocorrelation function* is defined in the obvious way, as

$$\hat{\rho}_X(h) = \frac{\hat{\gamma}_X(h)}{\hat{\gamma}_X(0)}.$$

We typically inspect the sample autocorrelation function, shortened to the ACF, when looking at time series.

The sample autocovariance is an asymptotically unbiased estimate of the autocovariance, that is, for fixed h ,

$$\lim_{n \rightarrow \infty} E(\hat{\gamma}_X(h)) = \gamma_X(h).$$

- 1 The sample autocovariance and autocorrelation functions inherit the
2 properties possessed by the population versions.
- 3 In view of these properties, we plot the sample autocorrelation func-
4 tion like shown in Figure 4.

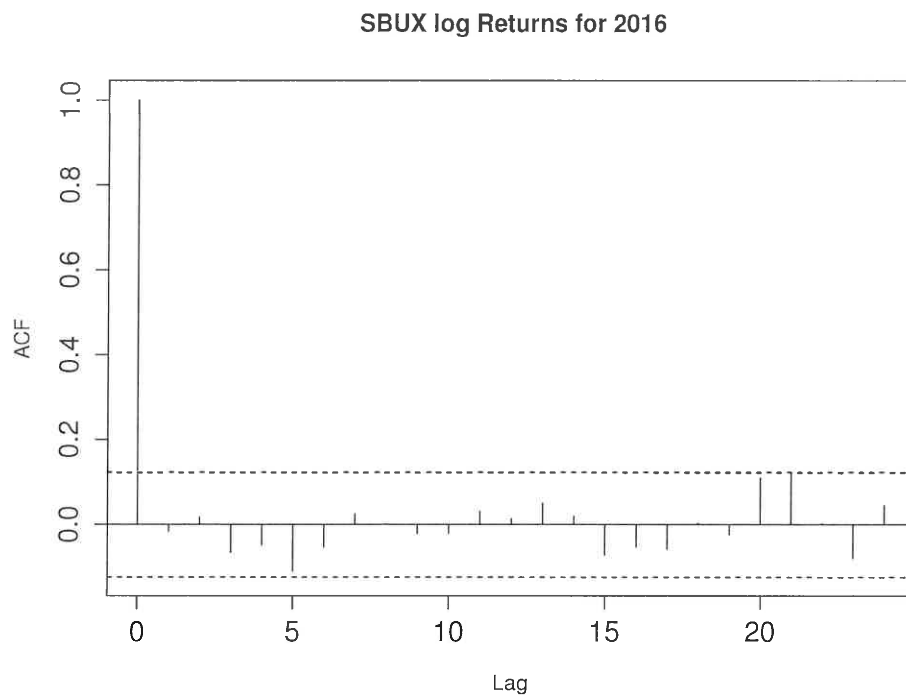


Figure 4: Sample ACF of the log returns for SBUX for 2016.

- 5 Note that since it is symmetric, we don't bother to plot it for negative
6 values of h .
- 7 The function `acf()` in R can plot a sample ACF (and a sample auto-
8 covariance function).

1 The Ljung-Box Test

2 Figure 4 shows dashed lines representing significance bounds on the
3 individual estimates of $\rho_X(h)$. The interpretation is as follows:

4 If $\hat{\rho}_X(h)$ falls within these bounds, there is insufficient evidence to
5 reject $H_0: \rho_X(h) = 0$ in favor of $H_1: \rho_X(h) \neq 0$.

6 One should keep in mind that these are **tests on the individual coef-**
7 **ficients**.

8 **Exercise:** Explain why one should be careful when interpreting the
9 bounds as shown in Figure 4.

10 Multiple testing problem

11 If I test n hypotheses and in fact the
12 null is true in every case, using $\alpha = 0.05$,
13 then expect to incorrectly reject $0.05n$
14 of them

15 In example, expect $(0.05)(25) > 1$ to be
16 outside the bounds, even if $\rho_X(h) = 0$
17 for all h
18

1 **R Exercise:** Simulate white noise, and then calculate its ACF. Do this
2 repeatedly. How often do any of the values of $\hat{\rho}_X(h)$ fall outside of
3 the given confidence bounds? What are the implications of this?

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1 **Exercise:** Explain the appeal of testing

2
$$H_0: \rho_X(h) = 0 \text{ for all of } h = 1, 2, \dots, K$$

3 versus

4
$$H_1: \rho_X(h) \neq 0 \text{ for some } h = 1, 2, \dots, K.$$

5
6 If I reject H_0 , have strong evidence
7 of $\rho_X(h) \neq 0$ for some h
8
9
10
11

12 In order to test the null hypothesis that **all** of the $\rho_X(h) = 0$, one can
13 utilize the *Ljung-Box Test*.

- 1 This is implemented in R as `Box.test()` with `type="Ljung-Box"`.
- 2 To adjust K , set the argument `lag`.

```
3 > Box.test(dailyReturn(Ad(SBUX)), lag=20,  
4           type="Ljung-Box")  
5  
6 Box-Ljung test  
7  
8 data:  logret  
9 X-squared = 8.782, df = 20, p-value = 0.9853
```

10 **Exercise:** Interpret the results of the above test.

11 p-value of 0.9853 suggests no evidence
12 of correlation
13
14
15
16

Partial Autocorrelation Function

Figure 4 shows little correlation at lags 1 to 20; this is probably not terribly surprising.

It can be sometimes be of interest to pose a slightly different question:

Is there strong correlation between X_t and X_{t-k} **conditional on** the intervening values $X_{t-1}, X_{t-2}, \dots, X_{t-k+1}$?

In other words, it is like asking: Is there a strong correlation between X_t and X_{t-k} even after we've adjusted for all of the information that occurred between times t and $t - k$?

This is explored via the *partial autocorrelation function*.

1 Imagine fitting a regression model of the form

2
$$X_t = \phi_{0,k} + \phi_{1,k}X_{t-1} + \cdots + \underbrace{\phi_{k,k}}_{\text{sample partial autocorrelation}}X_{t-k} + \epsilon_{k,t}$$

3 where the ϕ are the parameters to be estimated. Then $\hat{\phi}_{k,k}$ is the k^{th}
4 sample partial autocorrelation.

5 One imagines refitting this model for each value of k to obtain $\hat{\phi}_{1,1}, \hat{\phi}_{2,2}, \dots$
6 and then these values are plotted.

7 In R, this is accomplished using the function `pacf()`. See the result
8 on our SBUX data as Figure 5.

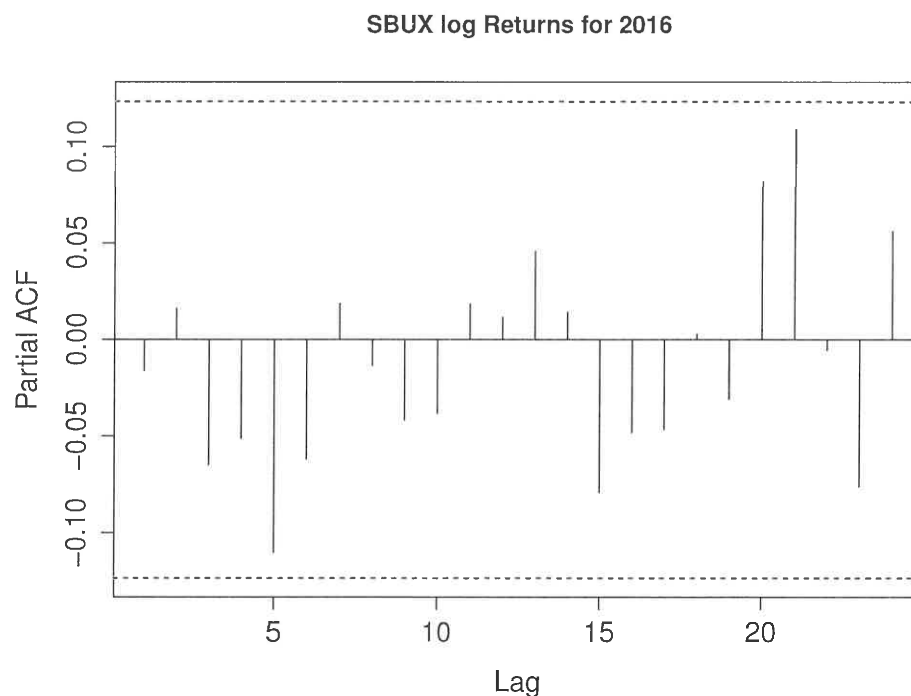


Figure 5: Sample PACF of the log returns for SBUX for 2016.

AR(1)

$$X_t = \mu + \beta X_{t-1} + \epsilon_t$$

Conditioned on X_{t-1} , X_{t-2} and X_t are uncorrelated,

PACF at lag 2 is 0

But ACF at lag 2 is not zero

Quantmod and XTS Functions

Here we will go through some of the functions included in the R packages `quantmod` and `xts`.

The standard R distribution has time series functionality built in, but `xts` adds on a great deal of flexibility in working with financial data. In particular, `xts` allows one to work naturally within the trading calendar.

The objects which are produced by `getSymbols()` are in the `xts` format. We will use that as a starting point, prior to constructing our own `xts` objects.

Assume we begin by creating an object of equity data:

```
> MSFT = getSymbols("MSFT", from="2010-1-1",  
                    to="2012-12-31", auto.assign=F)
```

Note that I generally prefer to use the `auto.assign=F` argument so that it does not automatically create objects.

1 Subsetting the Data

2 These functions extract particular columns:

3 > Op (MSFT)

4 > Hi (MSFT)

5 > Lo (MSFT)

6 > Vo (MSFT)

7 > Ad (MSFT)

8 The extremes of the data set:

9 > seriesHi (MSFT)

10 > seriesLo (MSFT)

11 Somewhat obvious functions:

12 > first (MSFT)

13 > last (MSFT)

14 > ndays (MSFT)

15 > nmonths (MSFT)

16 > nquarters (MSFT)

17 > nyyears (MSFT)

1 Only certain dates:

```
2 > MSFT['2012']
3 > MSFT['2012-01']
4 > MSFT['2012-01-01::2012-02-15']
5 > MSFT['2012-02::']
6 > MSFT[c('2011-01', '2012-01')]
```

7 Special subsets (note that these are **calendar** weeks/months/years):

```
8 > last(MSFT, '2 weeks')
9 > last(MSFT, '-2 weeks')
10 > first(MSFT, '1 week')
11 > first(MSFT, '20 days')
```

12 Aggregation:

```
13 > to.weekly(MSFT)
14 > to.monthly(MSFT)
```

15 The endpoints of time periods:

```
16 > endpoints(MSFT, on="weeks")
17 > endpoints(MSFT, on="months")
18 > endpoints(MSFT, on="years")
```

1 **Calculating Returns**

2 The default is `type="arithmetic"`. You can also set `type="log"`.

3 Note that this uses the column `Close` of the data set.

4 `> dailyReturn(MSFT)`

5 `> weeklyReturn(MSFT)`

6 `> quarterlyReturn(MSFT)`

7 `> yearlyReturn(MSFT)`

1 Some Plots

2 Take a look at each of the following to see what they create. In
3 each case there is ample opportunity to customize the output; see
4 the `help()` file for each.

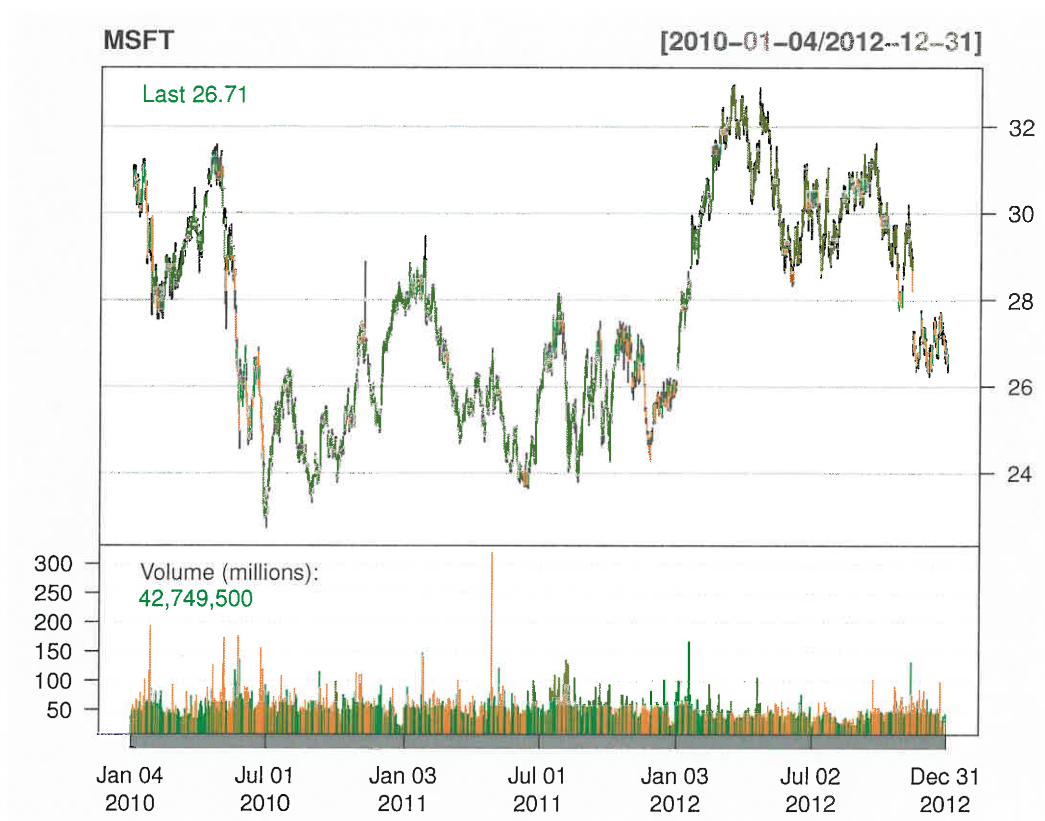
5 `> plot(MSFT)`

6 `> chartSeries(MSFT)`

7 `> barChart(MSFT)`

8 `> candleChart(MSFT)`

9 `> lineChart(MSFT)`



1 Creating a New XTS Object

2 A vector or data frame can be translated into an `xts` object by using
3 the function `xts()`. Keep in mind that the vector of dates must be
4 supplied; often, these are copied from another object.

5 Try the following toy code:

```
6 > x = rnorm(ndays(MSFT))  
7 > x = xts(x, time(MSFT))
```

8 Note that the `time()` function just returns the vector of dates for the
9 `xts` object `MSFT`.